

ETL Pipeline

~1.2 Billion Records Processed in a 5 GB Environment

Fannie Mae Single-Family Loan Performance Data
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What the pipeline does

- Processes 48 Fannie Mae quarterly loan performance files - over one billion pipe-delimited monthly records
- Operates inside a 5 GB SAS OnDemand environment with a 1 GB per-file upload limit
- Quarters exceeding the 1 GB upload limit are pre-split on loan boundaries by a PowerShell utility: Cutting only between loans keeps loan history intact
- Source files are read directly from gzip through the FILENAME ZIP engine: Enables 35 GB of raw text to be processed within the 5 GB storage limit
- Retains only the fields the scorecard requires, applies informats to convert MMYYYY dates and coded text, and derives the modelling fields: minimum credit score, origination quarter, and the default flag
- Collapses each loan history of monthly records into a single analytical row carrying its origination characteristics and final outcome, then appends it to the master table
- Runs idempotently - reprocessing a file replaces its prior load, never duplicates it
- Records the counts for every file in the processing record, which reconciles to Fannie Mae published statistical summaries

ETL Pipeline Processing Record

Sequence	Acquisition quarter	Source file	Monthly records read	Loans after collapse	Defaults identified	Processing time	Cumulative records
1	2000Q1	2000Q1.csv.gz	9,236,094	246,862	3,390	0:00:58.66	9,236,094
2	2000Q2	2000Q2.csv.gz	8,344,168	274,339	3,556	0:00:53.35	17,580,262
3	2000Q3	2000Q3.csv.gz	8,869,382	333,498	3,691	0:00:56.99	26,449,644
4	2000Q4	2000Q4.csv.gz	10,358,630	363,892	4,824	0:01:06.98	36,808,274
5	2001Q1	2001Q1.csv.gz	15,838,528	471,277	6,228	0:01:41.93	52,646,802
6	2001Q2	2001Q2.csv.gz	32,114,758	847,128	9,606	0:03:25.03	84,761,560
7	2001Q3	2001Q3.csv.gz	28,527,792	790,095	9,278	0:03:01.70	113,289,352
8	2001Q4	2001Q4.csv.gz	38,135,841	896,643	10,576	0:04:01.68	151,425,193
9	2002Q1	2002Q1.csv.gz	42,663,862	968,761	11,277	0:04:31.44	194,089,055
10	2002Q2	2002Q2.csv.gz	27,092,919	669,494	9,807	0:02:52.36	221,181,974
11	2002Q3	2002Q3.csv.gz	33,649,174	751,084	10,919	0:03:33.01	254,831,148
12	2002Q4	2002Q4_01.csv.gz	30,000,009	514,243	5,560	0:03:08.52	284,831,157
13	2002Q4	2002Q4_02.csv.gz	30,000,088	514,894	5,558	0:03:10.83	314,831,245
14	2002Q4	2002Q4_03.csv.gz	12,843,706	219,803	2,388	0:01:21.02	327,674,951
15	2003Q1	2003Q1_01.csv.gz	30,000,015	450,411	5,435	0:03:09.84	357,674,966
16	2003Q1	2003Q1_02.csv.gz	30,000,062	451,483	5,512	0:03:10.33	387,675,028
17	2003Q1	2003Q1_03.csv.gz	30,000,001	450,636	5,597	0:03:10.23	417,675,029
18	2003Q1	2003Q1_04.csv.gz	5,011,897	75,328	893	0:00:31.84	422,686,926
19	2003Q2	2003Q2_01.csv.gz	30,000,063	390,956	5,694	0:03:12.34	452,686,989
20	2003Q2	2003Q2_02.csv.gz	30,000,117	389,669	5,692	0:03:09.79	482,687,106
21	2003Q2	2003Q2_03.csv.gz	30,000,052	390,367	5,569	0:03:10.14	512,687,158
22	2003Q2	2003Q2_04.csv.gz	30,000,005	389,778	5,704	0:03:12.16	542,687,163
23	2003Q2	2003Q2_05.csv.gz	7,056,863	92,187	1,340	0:00:45.34	549,744,026
24	2003Q3	2003Q3_01.csv.gz	30,000,068	346,492	5,622	0:03:09.86	579,744,094
25	2003Q3	2003Q3_02.csv.gz	30,000,258	347,173	5,538	0:03:13.74	609,744,352
26	2003Q3	2003Q3_03.csv.gz	30,000,009	346,625	5,650	0:03:10.98	639,744,361
27	2003Q3	2003Q3_04.csv.gz	30,000,072	346,995	5,511	0:03:10.72	669,744,433
28	2003Q3	2003Q3_05.csv.gz	30,000,040	347,202	5,616	0:03:10.02	699,744,473
29	2003Q3	2003Q3_06.csv.gz	363,299	4,125	51	0:00:02.34	700,107,772
30	2003Q4	2003Q4_01.csv.gz	30,000,091	371,069	7,848	0:03:08.24	730,107,863
31	2003Q4	2003Q4_02.csv.gz	30,000,004	370,986	7,892	0:03:08.42	760,107,867

ETL Pipeline Processing Record

Sequence	Acquisition quarter	Source file	Monthly records read	Loans after collapse	Defaults identified	Processing time	Cumulative records
32	2003Q4	2003Q4_03.csv.gz	7,949,259	98,333	2,158	0:00:50.01	768,057,126
33	2004Q1	2004Q1.csv.gz	35,954,755	452,471	12,197	0:03:45.92	804,011,881
34	2004Q2	2004Q2.csv.gz	51,180,070	614,487	16,335	0:06:18.22	855,191,951
35	2004Q3	2004Q3.csv.gz	28,708,821	389,487	12,398	0:03:29.23	883,900,772
36	2004Q4	2004Q4.csv.gz	28,695,574	361,659	14,160	0:03:30.42	912,596,346
37	2005Q1	2005Q1.csv.gz	24,701,544	303,611	12,921	0:03:00.00	937,297,890
38	2005Q2	2005Q2.csv.gz	27,084,301	339,372	17,269	0:03:18.17	964,382,191
39	2005Q3	2005Q3.csv.gz	35,679,987	440,521	26,096	0:04:24.82	1,000,062,178
40	2005Q4	2005Q4.csv.gz	29,647,091	378,311	25,556	0:03:37.75	1,029,709,269
41	2006Q1	2006Q1.csv.gz	18,179,171	253,043	19,135	0:02:12.45	1,047,888,440
42	2006Q2	2006Q2.csv.gz	19,718,207	291,165	23,010	0:02:24.65	1,067,606,647
43	2006Q3	2006Q3.csv.gz	16,542,961	271,373	21,450	0:02:01.52	1,084,149,608
44	2006Q4	2006Q4.csv.gz	18,044,494	280,979	23,835	0:02:11.97	1,102,194,102
45	2007Q1	2007Q1.csv.gz	16,759,833	253,279	21,612	0:02:03.54	1,118,953,935
46	2007Q2	2007Q2.csv.gz	18,587,946	287,263	24,650	0:02:16.33	1,137,541,881
47	2007Q3	2007Q3.csv.gz	18,293,962	314,703	26,392	0:02:14.94	1,155,835,843
48	2007Q4	2007Q4.csv.gz	22,672,192	391,181	38,512	0:02:47.15	1,178,508,035

Data sets processed	Total monthly records	Total loans	Total defaults	Total SAS processing time	Records per second
48	1,178,508,035	19,144,733	519,508	2:10:57	149,997

